Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=96 (kcal/mol)
Predicted most stabilizing per doma
10
1
0
8.67
13.25
22.47
4.45
9.10
14.21
5.03
9.45
15.07
3.51
6.66
10.05
3.70
8.21
12.55
4.09
8.03
13.42
3.67
4.15
5.22
3.89
4.84
6.24
0.77
0.84
1.49
1.94
5.05
10.51
1.53
4.94
11.86
21.58
30.24
44.18
19.29
33.42
43.28
20.02
32.35
43.63
19.93
32.54
43.54
17.37
33.71
44.92
18.60
31.55
45.86
38.64
31.21
26.15
38.42
30.06
27.52
34.94
32.06
29.00
12.91
24.48
58.61
9.24
27.24
59.52
Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ volume ≤ −1
−1 < Δ volume ≤ 1
Δ volume > 1
Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,421 single mutants); gain = sum of max(ddG, 0) over the picks predicted stabilizing (> 0);
boxes: interquartile range with median line; dots: domains; annotations: mean over domains; dashed lines: oracle mean.